

Variability of Drop Size Distributions: Time-Scale Dependence of the Variability and Its Effects on Rain Estimation

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(Manuscript received 3 March 2004, in final form 16 July 2004)

ABSTRACT

A systematic and intensive analysis is performed on 5 yr of reliable disdrometric data (over 20 000 one-minute drop size distributions, DSDs) to investigate the variability of DSDs in the Montreal, Quebec, Canada, area. The scale dependence (climatological scale, day to day, within a day, between physical processes, and within a physical process) of the DSD variability and its effect on rainfall intensity R estimation from radar reflectivity Z are explored in terms of bias and random errors. Detail error distributions are also provided. The use of a climatological R - Z relationship for rainfall—affected by all of the DSDs' variability—leads on average to a random error of 41% in instantaneous rain-rate estimation. This error decreases with integration time, but the decrease becomes less pronounced for integration times longer than 2 h. Daily accumulations computed with the climatological R - Z relationship have a bias of 28% because of the day-to-day DSD variability. However, when daily R - Z relationships are used, a random error of 32% in instantaneous rain rate is still present because of the DSD variability within a day. This illustrates that most of the variability of DSDs has its origin within a storm or between storms within a day. Physical processes leading to the formation of DSDs are then classified according to the vertical structure of radar data as measured by a UHF profiler collocated with the disdrometer. The DSD variability among different physical processes is larger than the day-to-day variability. A bias of 41% in rain accumulations is due to the DSD variability between physical processes. Accurate rain-rate estimation ($\sim 7\%$) can be achieved only after the proper underlying physical process is identified and the associated R - Z relationship is used.

1. Introduction

The variability of drop size distributions (DSDs) determines the theoretical limit of the precision of precipitation measurements by radar. The practical interest of the radar reflectivity Z to rainfall intensity R transformation has been an important issue for over half of a century (Marshall and Palmer 1948; Joss and Waldvogel 1970; Richards and Crozier 1983; Willis 1984; Tokay and Short 1996; Smith and Joss 1997; Doelling et al. 1998; Uijlenhoet et al. 2003a,b; Lee et al. 2004). Nevertheless, in most operational radars, a simple relationship (such as $Z = 200R^{1.6}$ or $Z = 300R^{1.5}$) is used.

From a more fundamental point of view, the study of the microphysical processes responsible for the formation of DSDs and their evolution is an interesting but difficult subject because of the great variability of the possible processes involved. Many studies have been conducted to understand and describe key processes,

mainly the warm rain processes, such as autoconversion of cloud droplets to precipitation, coalescence, aggregation, breakup, and so on (Srivastava 1967; Young 1975; Passarelli 1978; Passarelli and Srivastava 1979; Valdez and Young 1985; Mitchell 1988; Rogers et al. 1991; Zawadzki et al. 1994; Hu and Srivastava 1995, to name just a few).

The connection between the two approaches, the practical and the fundamental, links remote sensing observations and theoretical considerations, providing a full description of the variability of DSDs. Fujiwara (1965) demonstrated that the systematic change of DSDs implies different R - Z relationships between precipitation types (continuous rains, rainshowers, and thunderstorm). Furthermore, he showed that this systematic difference is independent of geographical locations (Florida and Illinois). Sempere-Torres et al. (1999) and Lee (2003) attempt the classification of DSDs according to the dominant physical processes as revealed by radar observations and connect the classified dominant physical processes with R - Z relationships. Here we present results of a systematic classification on DSD variability, its scale dependence, its dependence on physical processes, and the implications on moment estimation.

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Measured DSDs suffer from several uncertainties such as the instrumental noise related to small sampling volume and observational noise due to the drop sorting effect. Even with a physically homogeneous period, these uncertainties can lead to R - Z relationships that are only true in a statistical sense. After filtering out the noise, a derived R - Z relationship may converge to a deterministic one relevant for the period. In addition, the R - Z relationship is not a deterministic relationship when all DSDs from various physical processes are combined. The most common rain estimation algorithm in operational radars still uses a single R - Z relationship. Hence, the scatter with respect to an R - Z relationship will determine the theoretical limit in rain estimation due to the DSD variability. Several investigators reported that the DSD variability explains 30%–50% of errors in R estimation with a single R - Z relationship (Goddard and Cherry 1984; Direskeneli et al. 1986; Balakrishnan et al. 1989) although their datasets were quite limited and were obtained from different geographical locations (i.e., different climate regions). This error will depend on the different integration time and on the R - Z relationships used. For example, in a long-term integration (i.e., 1 yr) the scat-

ter becomes negligible and Z and R will fit with the climatological R - Z relationship. On the other hand, it is important to notice that the error due to the variability of DSDs is a combination of random and bias errors. (Here the random error does not mean a complete uniformity in the distribution with respect to a zero mean. The skewness will be investigated in the appendix.) With a climatological R - Z relationship, the variability of DSDs gives the random error in instantaneous rainfall or short-term accumulation (i.e., hourly) at a given radar pixel (i.e., a few kilometers squared) during long periods such as a few months or years. This random error can be reduced by mixing rainfall intensities that originated from various physical processes. We can expect this mixing in long-term rain accumulation, such as monthly or yearly. Since an R - Z relationship for a particular physical process can be quite different from a climatological relationship, there is a significant amount of bias in addition to the random error when the rain accumulation or intensity is estimated for an hour or a few minutes with a climatological relationship.

An illustration of the bias and random errors is shown in Fig. 1. These data were taken on 1 May 1994

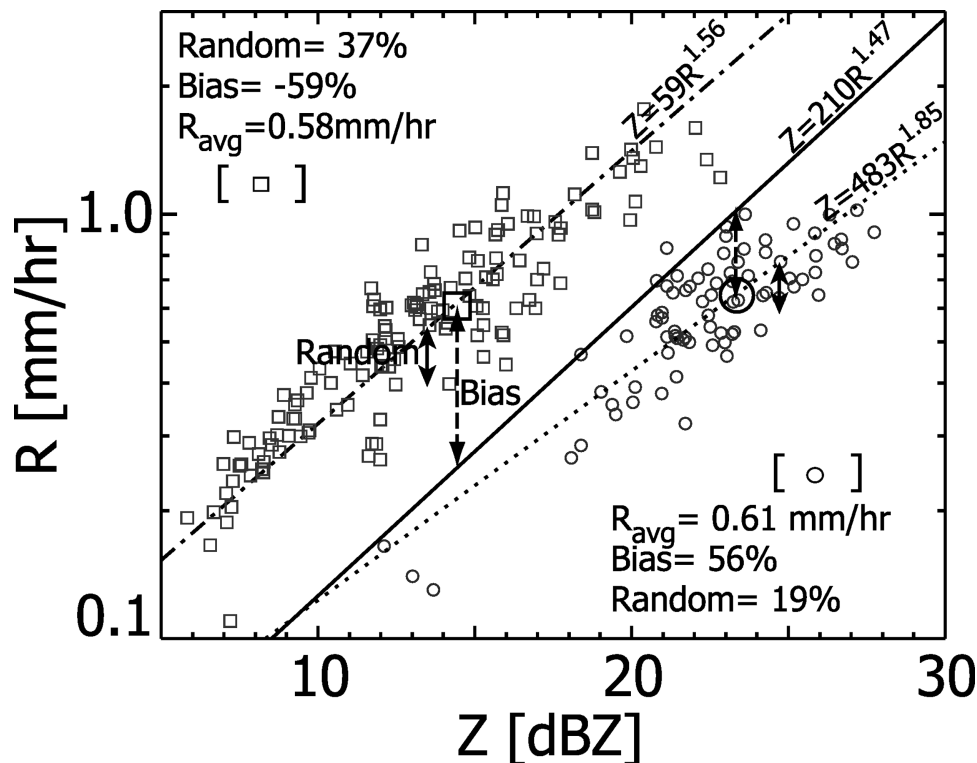


FIG. 1. Illustration of the bias and random errors for two physical processes, drizzle (squares) and melted snow (circles), on 1 May 1994. Each data point (small symbol) represents a 1-min DSD. The large symbols indicate the average for the processes. The best-fitted lines that are derived from R vs Z regression in log-log coordinates are shown. The solid line indicates the climatological R - Z relationship. The bias is derived from $(R_{T,avg} - R_{avg})/R_{avg}$, where R_T is calculated with the climatological R - Z relationship and "avg" indicates an average over the physically homogeneous period. The random error is derived from $[(1 - R_T/R)^2]_{avg}^{1/2}$, where the best-fitted lines are used to calculate R_T .

and the event can be divided into two distinct types of precipitation: drizzle and rain with large drops resulting from melting snow. The best-fitted equations for the two periods are different from the climatological $Z = 210R^{1.47}$. For the drizzle case (squares in Fig. 1), most data points lie above the climatological line and vice versa for the rain period (circles). Hence, when rainfall intensities are estimated with the climatological relationship, a bias of -59% $[(0.24-0.58)/0.58]$ and 56% $[(0.95-0.61)/0.61]$ is expected for the drizzle and rain cases, respectively. In addition to the bias, there exists a scatter around the best-fitted lines. Thus, random errors of 37% and 19% are expected for instantaneous R . This random error is due to a combination of “measurement and instrumental noise” and nonhomogeneity of the physical processes (see Lee and Zawadzki 2003). We will investigate the variability of DSDs at different scales (such as climatological scale, day to day, and between physical processes) and its effects on rain estimation in terms of the bias and random errors to provide a baseline for the estimation of rain from Z .

In section 2, we investigate the climatological variability and its effects on rain estimation. The detailed error distribution is also shown in the appendix. The day-to-day variability and variability within a day are presented in section 3. Furthermore, in section 4, we show that a classification of physical processes is necessary in order to reduce the errors in rain estimation due to the DSD variability. We summarize our results in the section 5.

The data used here are from 60 days of disdrometric observation, consisting of about 20 400 one-minute DSDs, taken over 5 yr (1994, 1997–2000) near Montreal, Quebec, Canada. The instrument is the Precipitation Occurrence Sensor System (POSS) described by Sheppard (1990). It measures an average Doppler spectrum and converts this spectrum into a DSD every minute. The sampling volume of POSS (Table 1) is three orders of magnitude larger than that of a conventional disdrometer [POSS: $V(D) = 0.3\text{--}190 \text{ m}^3 \text{ s}^{-1}$ depending on diameter; Joss–Waldvogel disdrometer: $V(D) = 0.007\text{--}0.05 \text{ m}^3 \text{ s}^{-1}$]. Thus, measured DSDs are less affected by the undersampling uncertainty. In addition, when the total number of drops at a given diameter for a minute is less than 10 drops, we discard DSDs at this diameter in all data analysis to avoid the undersampling effect. Sheppard and Joe (1994) and Campos and Zawadzki (2000) have provided a detailed validation of POSS as a disdrometer by comparison with other instruments.

2. Variability of DSDs at climatological scale

The variability of DSDs depends on the space and time scales over which they are observed. In instantaneous radar measurements, we are interested in the variability of the DSD averaged over a radar pixel ($0.1\text{--}1 \text{ km}^3$). This variability also depends on the time of integration (i.e., instantaneous, hourly, or daily aver-

TABLE 1. Mean diameter, diameter interval, and measurement volume of POSS data.

	Mean diameter, D (mm)	Diameter interval, dD (mm)	Measurement volume, $V(D)$ ($\text{m}^3 \text{ s}^{-1}$)
1	0.34	0.05	0.32
2	0.38	0.05	0.44
3	0.44	0.05	0.71
4	0.49	0.05	0.78
5	0.54	0.06	1.0
6	0.60	0.06	1.3
7	0.66	0.06	1.9
8	0.72	0.06	2.4
9	0.78	0.06	2.9
10	0.84	0.06	3.7
11	0.91	0.07	4.3
12	0.97	0.07	5.2
13	1.05	0.07	5.8
14	1.12	0.08	6.9
15	1.20	0.08	7.9
16	1.28	0.08	8.6
17	1.37	0.09	10
18	1.46	0.09	11
19	1.55	0.10	13
20	1.65	0.10	15
21	1.76	0.11	18
22	1.87	0.12	21
23	2.00	0.12	26
24	2.12	0.13	30
25	2.26	0.14	36
26	2.40	0.15	41
27	2.56	0.17	49
28	2.73	0.18	58
29	2.92	0.20	73
30	3.14	0.23	92
31	3.40	0.28	110
32	3.70	0.36	130
33	4.15	0.56	150
34	5.34	1.84	190

age). In hydrology, the size of the subbasin and a 1-h interval are the typical scales of interest. The variability of DSDs causes an error in radar rain estimation since it is responsible for different $R\text{--}Z$ relationships. The magnitude of the error, hence, depends on the $R\text{--}Z$ relationships used. In this section, we quantify the variability of DSDs at climatological scale and its effect on rain estimation.

a. Climatological variability

To derive the climatological variability, we use the entire dataset just described. Figure 2a shows the $R\text{--}Z$ scatterplot of 1-min observations. The rainfall intensity R (mm h^{-1}) and radar reflectivity Z ($\text{mm}^6 \text{ m}^{-3}$ or dBZ in $10 \log Z$) are derived from 1-min DSDs using the following equations:

$$R = 6\pi \times 10^{-4} \int_{D_{\min}}^{D_{\max}} v(D) D^3 N(D) dD \quad \text{and} \quad (1)$$

$$Z = \int_{D_{\min}}^{D_{\max}} D^6 N(D) dD, \quad (2)$$

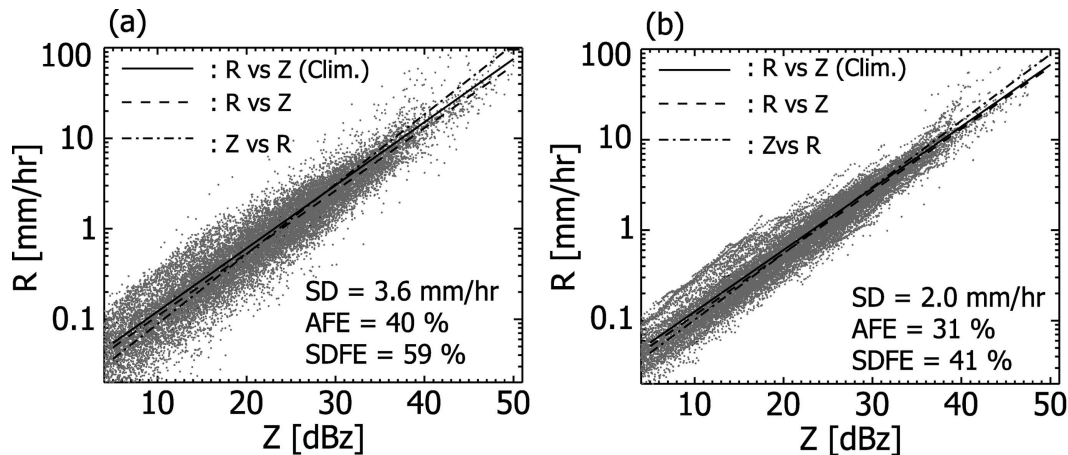


FIG. 2. (a) Scattergram derived from all the 1-min DSDs data. Several R - Z relationships are described in the text. (b) As in (a) but after applying SIFT for each day ($W = 1$ h and $M = 10$ samples of 1-min DSDs). Several statistics are derived with respect to the climatological relationship $Z = 210R^{1.47}$ (Fig. 3). Data with reflectivity less than 5 dBZ are not considered.

where $v(D)$ (m s^{-1}) is the terminal fall velocity according to Gunn and Kinzer (1949) and $N(D)$ ($\text{m}^{-3} \text{mm}^{-1}$) is the number of drops in a unit volume and given diameter interval dD (mm) centered at D (mm). The degree of scatter is quite large. In addition to the physical variability, the observed DSDs include the spurious variability due to the drop sorting and undersampling (Smith et al. 1993; Joss and Zawadzki 1997; Chandrasekar and Bringi 1987; Lee 2003; Lee and Zawadzki 2003). Hence, the scatter expresses all variability of DSDs, including the noise in the observations and that due to the various microphysical processes, from storm to storm and within each storm. A least square regression of R versus Z on log-log coordinates yields $Z = 231R^{1.54}$, and the Z versus R regression leads to $Z = 228R^{1.35}$. As illustrated in Lee and Zawadzki (2003), in the presence of the scatter, the derived relationships depend on the regression methods, such as Z versus R or R versus Z .

The spurious variability of disdrometric measurements resulting from instrumental noise, poor sampling, and drop sorting is undesirable. To eliminate the spurious variability while keeping a physical variability, we apply a *sequential intensity filtering technique* (SIFT) described in Lee and Zawadzki (2003). SIFT is applied for each day within a moving time window of typically 1 h ($W = 1$ h) within which DSDs are ordered by increasing Z and an arithmetic average of 10 consecutive DSDs is taken ($M = 10$ samples of 1-min DSDs). The increase in our ability to discriminate between different R - Z relationships and DSDs as a result of applying SIFT is quite spectacular, as discussed in Lee and Zawadzki (2003). Lee and Zawadzki (2005) provide a thorough discussion of SIFT. After applying SIFT to the data in Fig. 2a, the scatter is significantly reduced (Fig. 2b). However, a considerable variability still remains. The derived relationship is $Z = 234R^{1.46}$ ($Z =$

$224R^{1.36}$) for an R versus Z (Z versus R) regression. Because of the variability, the exponent is slightly different.

We now average all DSDs within a number of consecutive reflectivity values. In other words, we apply SIFT with a window size of 5 yr and perform an arithmetic average of all DSDs within 5-dBZ intervals. This procedure is equivalent to the way that Marshall and Palmer (1948) derived their exponential distribution. This procedure filters out most of the variability and maintains only the climatological average physical process. It also reveals systematic changes in DSDs with rain intensities. The linear regression of R versus Z through the resulting pair of (R , Z) values computed from average DSDs leads to the relationship $Z = 210R^{1.47}$ (Fig. 3). From now on, we consider it the climatological R - Z relationship. This relationship is similar to the result obtained by Marshall and Palmer (1948) over 50 years ago using a primitive measurement technique. Furthermore, it is different from the four other previously obtained relationships in Fig. 2; which one of these relationships is to be used depends on the application. For example, the relationship derived from the R versus Z regression is preferable when we are interested in an estimation of R with the smallest errors. When the physical processes as revealed by functional relationships are the main concern, the weighted total least square (WTLS; Amemiya 1997), which does not discriminate between dependent and independent variable, is preferred. In WTLS, scatter in both Z and R are minimized simultaneously. For a thorough discussion on WTLS and the uncertainty due to the regression methods, see Lee and Zawadzki (2003, 2005). However, after applying SIFT, the relationship from various regression methods converges. Henceforth, all R - Z relationships will be calculated here by regression of R versus Z and after applying SIFT.

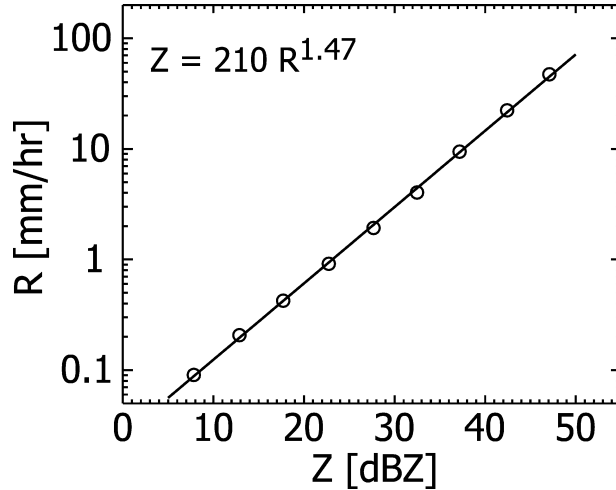


FIG. 3. The climatological R - Z relationship from DSDs obtained by averaging all 1-min data within 5-dBZ intervals of the reflectivity factor. The first interval is 5–10 dBZ and the last is 45–50 dBZ.

Now we calculate the error in rain estimation when R is derived from Z using the climatological R - Z relationship ($Z = 210R^{1.47}$). The statistics are shown in Fig. 2a. The standard deviation (SD), the average fractional error (AFE), and the standard deviation of fractional error (SDFE) are calculated with the following equations:

$$\begin{aligned} SD &= \left[\frac{1}{k} \sum (R - R_T)^2 \right]^{1/2}, \\ AFE &= \frac{1}{k} \left(\sum \frac{|R - R_T|}{R} \right), \text{ and} \\ SDFE &= \left[\frac{1}{k} \sum \left(\frac{R - R_T}{R} \right)^2 \right]^{1/2}, \end{aligned} \quad (3)$$

where k is the number of data points and R is the rainfall rate calculated from the 1-min DSDs using (1). Here, R_T is the transformed rainfall rate from the calculated reflectivity of the 1-min DSDs with (2) using our climatological R - Z relationship. The same error analysis is performed for half of the dataset selected randomly. Results are consistent with those from the whole dataset. This indicates that our sample size is sufficiently large to yield robust results.

When all variability is maintained, including the noise in the observations and the mixing of all the physical processes, AFE = 40% and SDFE = 59%. After eliminating most of the observational noise in Fig. 2b, AFE = 31% and SDFE = 41%. The latter are more representative of radar observations and may thus be considered as the theoretical limits of instantaneous rain estimation from radar with a single R - Z relationship. These are the random errors for 1-min measurements during long periods, which are expected

to decrease with integration time by mixing rain intensities originating from various physical processes (to be shown in section 2b). Detailed error distributions at climatological scale are shown in the appendix.

The magnitude of the error (SDFE = 41%) is within the range of values (30%–50%) obtained by other investigators (Goddard and Cherry 1984; Direskeneli et al. 1986; Balakrishnan et al. 1989) at different geographical locations (Europe, Colorado, and Oklahoma) although they derived the error from very limited datasets. The convective storms in the Great Plains of the United States are often associated with graupel and hail, which are seldom found in Montreal. Thus, comparable statistics do not necessarily imply similar rain processes.

b. Rain estimation errors at various integration times

Now, we study how the error in R estimates from Z , caused by the DSD variability, varies as a function of integration time (Δt). We expect the rain estimation error to decrease with integration time due to the smoothing out of the variability in the R - Z relationship (whether it is the physically based variability or the one caused by random errors in disdrometer measurements).

For this purpose, we select all precipitation events lasting more than 2 h from the dataset. For each event, DSDs are averaged over a running mean window of variable duration (Δt). Values of average reflectivity $\bar{Z}$ and average rain rate $\bar{R}$ are computed from the averaged DSDs using (2) and (1), respectively, and $\bar{R}$ - $\bar{Z}$ relationships are derived for each Δt . Error statistics are computed by comparing the actual values of $\bar{R}$ with the ones derived from the $\bar{R}$ - $\bar{Z}$ relationships. When $\Delta t > 2$ h, the number of independent cases decreases considerably from 154 with $\Delta t = 2$ h to 60 with $\Delta t = 6$ h. The results are shown in Fig. 4.

The standard deviation rapidly decreases as a function of integration time up to 2 h. By averaging 10 samples of 1-min DSDs ($\Delta t = 10$ min), $SD = 1.7$ mm h^{-1} (one-half of a standard deviation at averaging size = 1). AFE and SDFE also drastically decrease. This reduction of errors is less pronounced for an averaging period larger than 1 h (note that the x axis is a logarithmic coordinate). Since the variability of the raw DSDs is a combination of disdrometric measurement noise and physical DSD variability, the reduction of errors is a consequence of smoothing out of the two effects. The decrease in error is quite steady up to $\Delta t = 2$ h. No discernable break separates instrumental noise from the DSD variability.

The errors stabilize at $\Delta t > 2$ h ($SD = 0.7$ mm h^{-1} and $SDFE \sim 30\%$). It is interesting that further time averaging cannot reduce the errors. Since all errors are derived using a unique $\bar{R}$ - $\bar{Z}$ relationship appropriate for a given time scale, this result is due to the significant scatter remaining around such a relationship at time

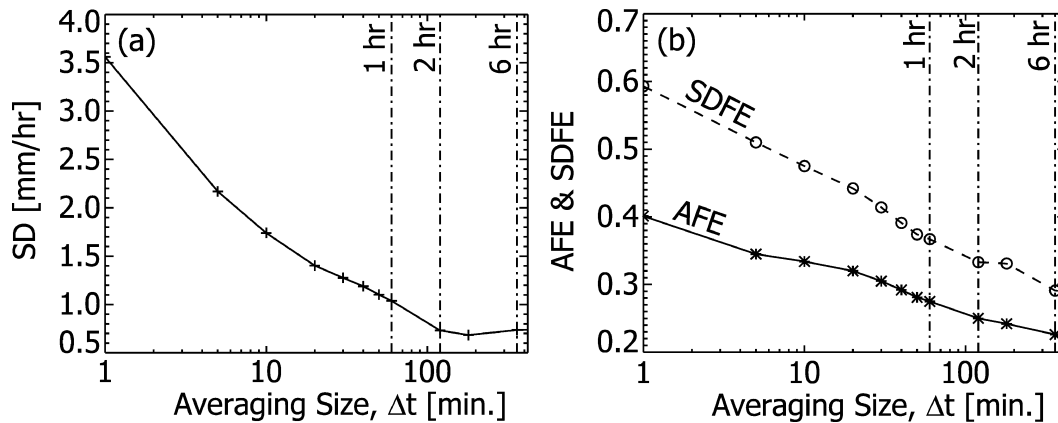


FIG. 4. Errors in estimation of average rain intensities at various integration times (from 1 min to 6 h): (a) the std dev and (b) the average fraction error (AFE) and std dev of fractional error (SDFE). For the statistics, raw data with values less than 10 dBZ are excluded.

scales over 2 h, indicating that the dominant physical process within this time scale is variable and far from the average dominant physical process of the entire storms.

3. Day-to-day variability and variability within a day

a. Day-to-day variability

By applying SIFT to the entire dataset of DSDs (leading to Fig. 3), that is, with a window of 5 yr and averaging of DSDs within 5-dBZ intervals, we have eliminated all variability of DSDs within a storm, from storm to storm, and between days. On the other hand, by applying SIFT with $W = 1$ h and $M = 10$ DSDs (Fig. 2b), we maintain all variability at time scales larger than 1 h and even some variability within 1 h. In what follows

we will narrow our interests down to the day-to-day variability of DSDs and of R - Z relationships.

We first examine the variability of daily R - Z relationships. Since daily R - Z relationships are related to daily dominant microphysical processes generating DSDs, this will provide an idea of the variation of dominant processes between days. SIFT is applied to daily datasets of DSDs with $W = 1$ day and $M = 2$ -dBZ interval. That is, all the DSDs within intervals of 2 dBZ observed in a given day are averaged. Thus, the only DSD variability that is retained is the one occurring from one reflectivity interval to the next. From these averaged DSDs we derive the daily R - Z relationships, which are shown in Fig. 5a. The length of the lines corresponds to the range of the reflectivity and of rain-fall rate in any given day. Since the day-to-day variability of the R - Z relationships is quite wide, the use of a single R - Z relationship would cause a significant daily

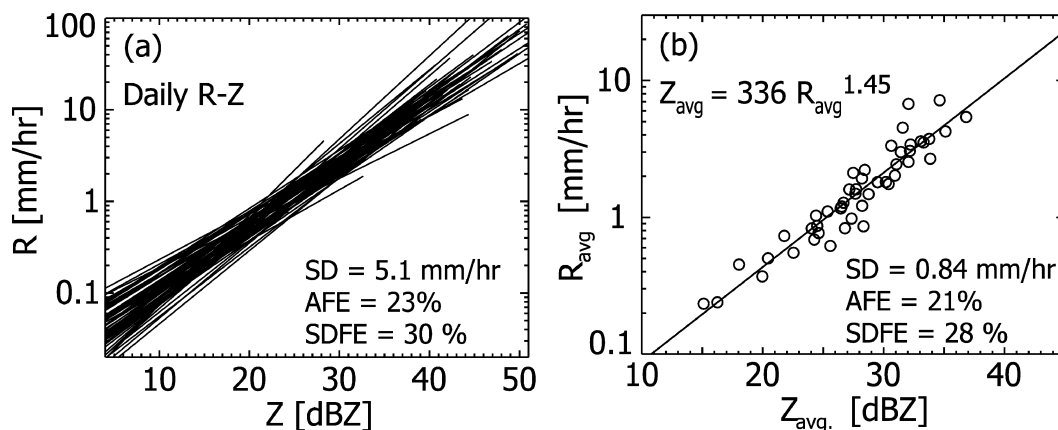


FIG. 5. (a) Daily R - Z regression lines obtained after SIFT is applied to daily data. The best fit ($Z = 248R^{1.45}$) of all these R - Z regression lines is derived and the statistics are obtained by comparing transformed rain rates from 2-dBZ reflectivity intervals with this best fit and with daily R - Z regressions. (b) Daily average R vs daily average Z . The best-fit line is $Z_{avg} = 336R_{avg}^{1.45}$. The statistics are derived with respect to the best fit.

bias. This also indicates that the dominant physical processes responsible for the generation of the DSDs are significantly different from day to day. We can quantify this difference by comparing R transformed from the daily R - Z relationships with R obtained from the best-fitted R - Z relationship of all the daily R - Z relationships. We thus obtain the bias in the estimation of daily rain accumulations with a single R - Z relationship instead of the daily R - Z relationships shown in Fig. 5a. The best-fitted relationship $Z = 248R^{1.45}$ is obtained from the scatter points taken from the daily R - Z lines at 2-dBZ reflectivity intervals. Then, the statistics are calculated by comparing the transformed R from 2-dBZ reflectivity intervals using the best-fitted relationships. A bias of 30% is expected due to the change of daily R - Z relationships. This error is slightly less than the random error in the estimation of instantaneous rain rate at the climatological scale in Fig. 2b.

The day-to-day variability can be investigated by considering daily averaged DSDs (or daily averaged R and Z). This completely filters out the variability within a day and maintains only the day-to-day variability. For each day, an averaged DSD is calculated, yielding daily average R and Z . Results are shown in Fig. 5b where each circle represents 1 of 60 days. A daily average DSD represents the dominant mean microphysics of the day. For example, let us assume that we have two physical processes within a day. Then, a daily average DSD should reflect which of the two is the dominant process. When the average DSD is wider than Marshall-Palmer (M-P) DSD, that is, a small number of drops at smaller sizes and a larger number density at the bigger sizes, we can speculate that active drop growth such as aggregation or coalescence is a dominant process. In fact, bigger Z is expected for the same R . Thus, the physical variability between days generates scatter around the line as shown in the figure. The day-to-day variability is quite significant as shown in the variation of daily R - Z relationships. When a single R - Z relationship ($Z = 336R^{1.45}$) is used to transform daily average Z into average R , this variability yields SDFE = 28%. Note that this error is the bias in the estimation of daily average R or accumulation with a single R - Z relationship. The bias becomes negligible when the correct daily R - Z relationships shown in Fig. 5a are used (SDFE = 8%; not shown here).

Instead of transforming the daily average Z into daily average R , 1-min reflectivities can be converted into rain rates using the climatological R - Z relationship $Z = 210R^{1.47}$ and then the average of these rain rates can be compared with daily average R derived from daily average DSDs. This alternate method also gives results similar to those in Fig. 5b (see Fig. 1 in Lee and Zawadzki 2004). This is the usual method of deriving average R or the accumulation from radar measurements; that is, R is estimated from Z measured from each volume scan and then the estimated R is accumulated. Conversely, the previous method accumulates in-

dividual Z over a certain time period and then transforms the average Z into average R . When the measurement noise is considered, the previous method is preferred since the noise can be effectively filtered out through initial averaging.

b. Variability within a day

Because of the day-to-day variability, the bias in daily accumulation is unavoidable but it could be reduced with the choice of a proper daily R - Z relationships. Now we further reduce the time scale of interest. The variability within a day can be expressed by the scatter in R - Z derived from 1-min DSDs with respect to the daily R - Z relationship. With observational noise and all physical variability, AFE = 34% and SDFE = 46%. After eliminating observational noises with SIFT for a given day ($W = 1$ h and $M = 10$ DSDs), AFE = 24% and SDFE = 32%. These errors are random and represent the theoretical limits in instantaneous rain estimation within a day with a proper daily R - Z relationship. When we integrate rain in time from 1 min to a day, this random error becomes zero and only the bias of 8% remains. This bias is the SDFE in daily average R using the proper daily R - Z relationship (see section 3a).

Interesting facts appear by comparing the variability within a day (SDFE = 32%) with the climatological variability (SDFE = 41%) in Fig. 2b. There is only a relatively small improvement of 9%; that is, random errors obtained from a climatological R - Z relationship are of the same order as those obtained from daily R - Z relationships. This implies that a significant amount of the variability comes from within a day, from changes in physical processes occurring within a day that are responsible for the formation of the DSDs, from storm to storm, or within a storm. Therefore, further classification of physical processes is mandatory in order to reduce the random error in rain estimation.

4. Variability between different physical processes

In this section, we quantify the degree of the variability of DSDs caused by changes in physical processes and the variability of DSDs within a dominant physical process. By physical processes we mean the mechanisms, such as snow riming, snow aggregation, convection, drizzle, evaporation, and so on, that may affect the form of the drop size distribution. Events are classified into different processes using the vertical structure of the Doppler velocity and reflectivity from the McGill UHF wind profiler collocated with the disdrometer. The distinction of cold and warm rain processes (i.e., stratiform and convective systems) uses the brightband (BB) signature (Fabry and Zawadzki 1995). For stratiform rain, the specific cold rain processes such as riming and aggregation are further identified by the magnitude of the Doppler velocity and vertical gradient of reflectivity.

tivity above the BB (Bell 2000; Zawadzki et al. 2001). In addition to the microphysical processes, dynamical causes such as low-level growth are also a source of DSD variability. The transition of a hurricane to an extratropical cyclone is another distinct process that results in quite different DSDs. Drizzle and evaporation cases are also identified. The classification criteria and the discussion of the physical causes for the variability of DSDs are presented in Lee (2003).

The dataset consists of over 20 storms described in Table 2. They have been selected because they each have certain characteristic features that suggest a particular physical process that is dominant throughout the duration of the DSD record. The R - Z scatterplots from these storms (Fig. 6) show the systematic change of relationships among quasi-homogeneous processes. This systematic change is related to the bias in the R estimate as explained in Fig. 1. From these scatterplots, the dependence of the bias on R or Z for each period can be easily identified. Furthermore, the scatter within any quasi-homogeneous process is remarkably small, providing a deterministic relationship within that period. This small scatter implies small random errors within a physical process (the scatter will be quantified at the end of this section).

The variability between quasi-homogeneous processes can be examined by

- (i) comparing R - Z relationships between periods where the consistency of the morphology of the vertical structure of the radar echo suggests a quasi-homogeneous process;
- (ii) deriving the average DSDs of each quasi-homogeneous process from which R_{avg} and Z_{avg} are deduced—the scatter in the R_{avg} versus Z_{avg} diagram

with respect to a best-fit line indicates the variability; and

- (iii) comparing R_{avg} derived from the average DSDs with “average R_T ,” that is, the average of transformed rain rates from reflectivities calculated from 1-min DSDs (the climatological relationship $Z = 210R^{1.47}$ is used in the transformation).

The comparison as in method i above can be seen in the scattergram of a and b of $Z = aR^b$ as shown in Fig. 7a with the order (from No. 0 to No. 20) of classified cases in Table 2. The constants of the climatological $Z = 210R^{1.47}$ are indicated by the cross symbols in Fig. 7a. A wide distribution with respect to the climatological relationship underlines the large variability of DSDs between physical processes. Drizzle (case 2) has a low coefficient ($a = 67$). Convection (cases 11 and 20), extratropical transition (case 16), evaporation (case 12), and riming (case 10) have large coefficients ($a \sim 530$). In particular, a and b are both large for a case where strong snow aggregation lead to rain with large drops (case 0). In general, the correlation between values of a and b stratified according to physical processes is low ($r = -0.34$).

Results from method ii above are shown in Fig. 7b. The best fit ($Z_{\text{avg}} = 238R_{\text{avg}}^{1.43}$) is shown as the solid line. Each circle is calculated from an average DSD for each physically quasi-homogeneous period given in Table 2. The scatter with respect to the best-fit line reflects the variability of the average DSDs among the different physical situations. As expected, when the individual R - Z relationships are quite different from the climatological $Z = 210R^{1.47}$, their contribution to the scatter in Fig. 7b is large (see cases 2, 10, 12, and 20). Because of this variability, AFE = 32% in estimation of R_{avg} and

TABLE 2. List of the various dominant physical processes and of their characteristics.

No.	Periods (UTC)	Characteristics
0	0510–0630 UTC 1 May 1994	Stratiform with strong aggregation, before brightband (BB) jump
1	0730–1020 UTC 1 May 1994	Stratiform after BB jump
2	1021–1245 UTC 1 May 1994	Convective drizzle
3	1415–1511 UTC 9 Mar 1998	Weak convection
4	1512–1615 UTC 9 Mar 1998	Stratiform with aggregation
5	1616–1800 UTC 9 Mar 1998	Very weak bright band with riming
6	2340–0130 UTC 5–6 May 1998	Typical stratiform with weak aggregation
7	0659–0800 UTC 15 Jun 1998	Low-level growth, bimodal distribution
8	0801–0919 UTC 15 Jun 1998	Low-level growth with excess of small drops and convection extending up to BB
9	1115–1245 UTC 15 Jun 1998	Low-level growth and convection extending into above BB
10	0531–0632 UTC 29 Jul 1998	Weak BB with fluctuation, weak upward motion; heavy riming or wet growth
11	0633–0712 UTC 29 Jul 1998	Strong convection
12	0601–0616 UTC 24 Aug 1998	Well-defined BB with vertical motions and strong evaporation
13	0617–0650 UTC 24 Aug 1998	Vigorous convection with low-level growth
14	0651–0715 UTC 24 Aug 1998	Well-defined BB and weak low-level growth
15	2210–0302 UTC 16–17 Sep 1999	Extratropical transition and high BB
16	0303–0459 UTC 17 Sep 1999	Extratropical transition, and thick and strong BB
17	0500–0649 UTC 17 Sep 1999	Extratropical transition, thick, strong BB; gradual decrease of the height of BB
18	0650–1330 UTC 17 Sep 1999	Extratropical transition, low BB, and heavy riming or graupel
19	1331–2018 UTC 17 Sep 1999	Extratropical transition and low/weak BB, heavily rimed particles
20*	1936–2019 UTC 24 Jul 2001	Strong convection and equilibrium DSD

* These data were obtained from measurements taken in Iowa City, IA.

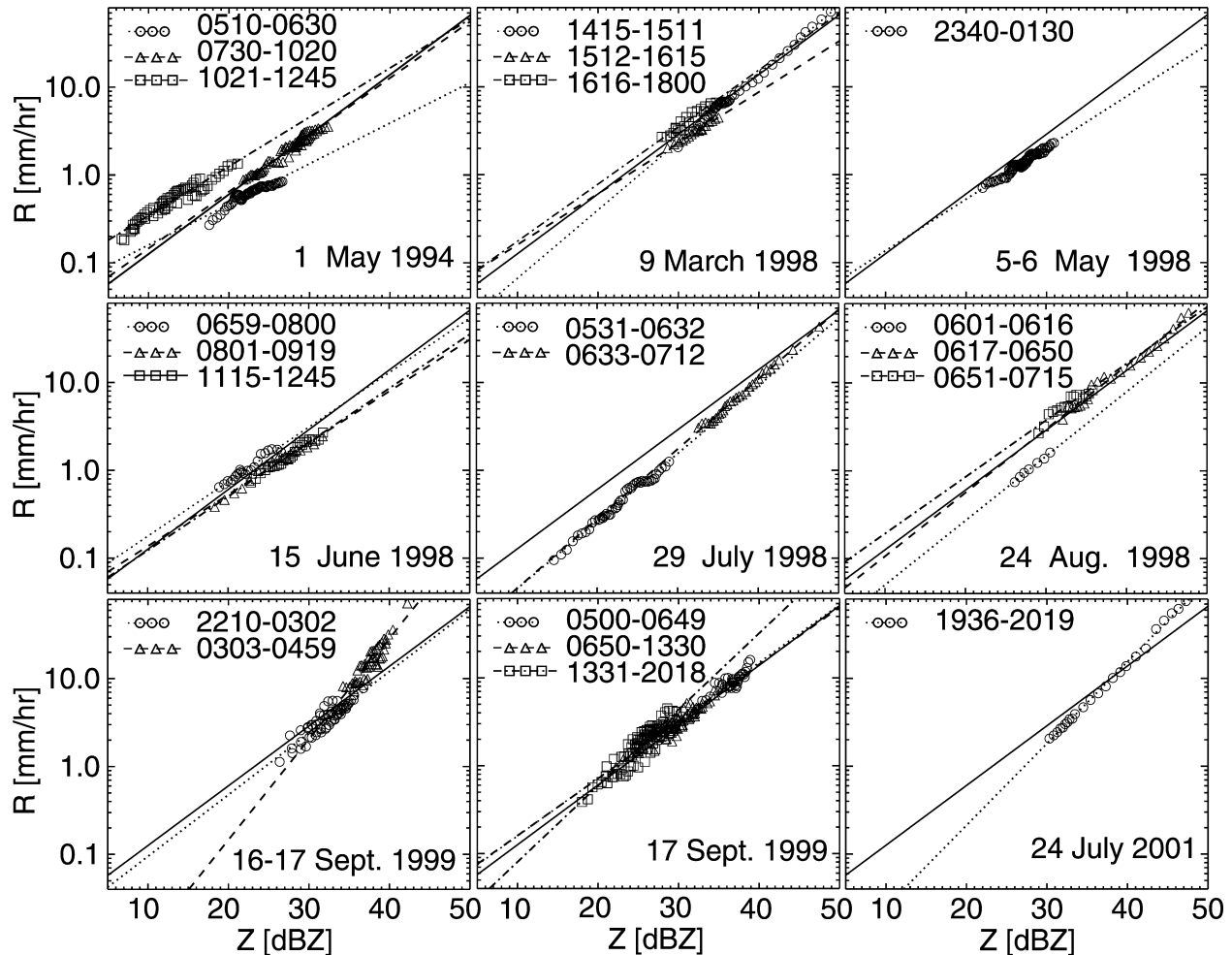


FIG. 6. The R - Z scatterplots for each physical process. For each period, SIFT is applied with $W=1$ h and $M=10$ one-minute DSDs. When the period is shorter than an hour, W is chosen as the length of the period. The solid line is the climatological relationship $Z = 210R^{1.47}$. The different lines are the best fits of R vs Z on log-log coordinate for given periods. For clarity in the figures, some data points (maximum 80%) are not displayed but there is no change in the overall features.

SDFE = 41%. It should be emphasized that these errors are *biases* that are expected when estimating rain accumulation in a case in which a particular physical process is dominant using a R - Z relationship ($Z_{\text{avg}} = 238R_{\text{avg}}^{1.43}$) valid on the average instead of using the R - Z that is appropriate for that particular process. Method iii above gives similar results (Fig. 7c): AFE = 34% and SDFE = 43%. The difference between them is marginal.

We can repeat the same procedure as in method iii above with the R - Z relationship for the physically quasi-homogeneous processes given in Fig. 7a. After using these R - Z relationships of the processes for the transformation (Fig. 7d), the error becomes negligible (SDFE = 1%).

The DSD variability between quasi-homogeneous physical processes is larger than the day-to-day variability (SDFE = 28% in Fig. 5b), because, for the day-to-day variability, different physical processes within a day

are combined, filtering out some of the variability. Because of this variability, a bias of 41% is expected in the estimation of rain accumulation for the physically quasi-homogeneous periods. The fact that the variability between physical processes is comparable with the climatological variability (SDFE = 41% in Fig. 2b) implies that most of it is a result of different physical processes. As described in the introduction (Fig. 1), the error due to the DSD variability is a combination of the bias and random errors. On the climatological scale, the error appears as purely random (41%). When we focus on the physical process, the total error (which should be equivalent to the climatological random error) is a combination of the bias originating from different physical processes and the random error within a physical process. As indicated by the small random error (7%; shown in the next paragraph), a combination of the bias and of random errors provides an equivalent value $[(41^2 + 7^2)^{1/2} = 42\%]$, leaving the bias between physical

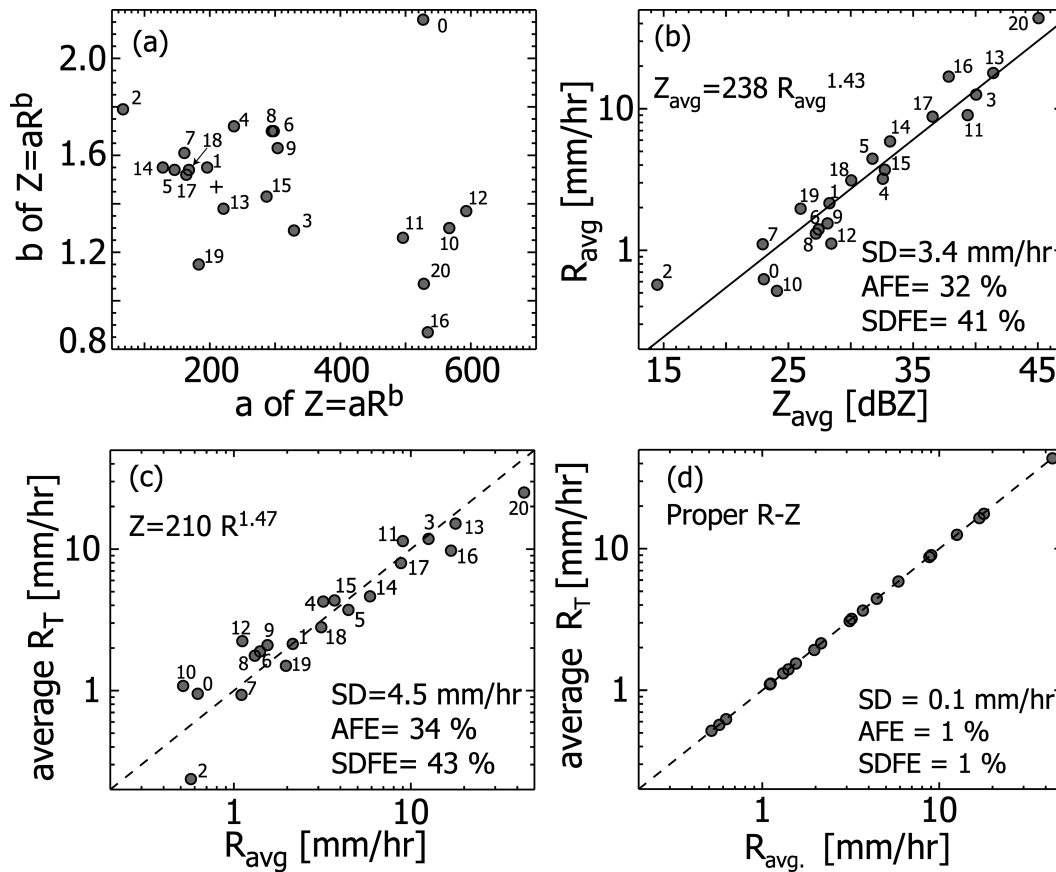


FIG. 7. (a) Scattergram of a and b of $Z = aR^b$ for the storm periods in Table 2. The constants of the climatological $Z = 210R^{1.47}$ are indicated by the cross symbol. The numbers next to the filled circles correspond to the order shown in Table 2. (b) Scattergram of average rain rate R_{avg} vs average reflectivity Z_{avg} for the physically quasi-homogeneous periods. The solid line is the best-fitted equation $Z_{avg} = 238R_{avg}^{1.43}$ for the R_{avg} and Z_{avg} dataset. The statistics are taken with respect to this equation. (c) Scattergram of averaged rain rates for the quasi-homogeneous periods obtained after transformation from Z to R vs actual average rates for the same periods. The relationship $Z = 210R^{1.47}$ is used for the transformation. (d) Same as in (c) but for the correct transformation for the process.

processes as the dominant factor. Thus we can conclude that most of the error is caused by the systematic change of DSD variability between physically homogeneous processes. Therefore, the classification or identification of different physical processes is essential in order to reduce the error in rain estimation. The vertical structure of precipitation can be well measured at near ranges with scanning radars. Thus, it is worthwhile to attempt the classification of precipitation and to make a connection between this classification and the variability of DSDs.

Similar to the analysis of the variability within a day, the variability within a quasi-homogeneous process can be analyzed from the scatter of 1-min reflectivities and rain rates with respect to a proper R - Z of the process, as shown in Fig. 6. This yields a random error of 7% (note the small scatter within a quasi-homogeneous period with respect to a proper R - Z). We can conclude that when the proper R - Z relationship for the process is used, the theoretical limit in the estimation of instan-

taneous rain rate is a random error of $\sim 7\%$ and a bias of $\sim 1\%$ (Fig. 7d). When integrating in time within a quasi-homogeneous process, the random error becomes zero and only the bias of 1% remains, the bias being the SDFE in average R within a quasi-homogeneous process using the proper R - Z relationship (Fig. 7d).

5. Summary and discussion

The variety of physical processes leading to the variability of DSDs is a key limitation in radar rain estimation. Numerous attempts have been made to reduce errors in rain estimation by better describing the variability of DSDs. Classification of precipitation systems and use of multiparameter measurements are examples of these attempts. In this paper, using a large sample of reliable disdrometric data, we investigate the variability of DSDs and the theoretical limit of rain estimation

from R - Z relationships at various time scales. The data used are disdrometric measurements taken by the Precipitation Occurrence Sensor System in Montreal over 5 yr. To eliminate observational and instrumental noise caused by the drop sorting and small sampling volume, the sequential intensity filtering technique is applied to temporal disdrometric data.

The variability of DSDs determines the theoretical limit in rain estimation. This limitation is investigated in terms of the random error and of the bias error as a function of various scales such as the climatological, day-to-day scales, and between physical processes. A summary of the results is shown in Table 3. Since the climatological variability is a mixture of all the physical variability, a random error of SDFE = 41% (Fig. 2b) in instantaneous rain estimation is expected when a single climatological R - Z relationship ($Z = 210R^{1.47}$) is used. This error is comparable to the results at different geographical regions obtained by other investigators (Godard and Cherry 1984; Direskeneli et al. 1986; Balakrishnan et al. 1989). This random error reduces with integration time but this reduction is less pronounced after 2 h of integration (Fig. 4) implying that time integration does not cause an average DSD to converge to a climatological DSD. A similar argument should also apply to the underlying physical processes in rain formation.

The detailed error distribution (the appendix) shows that the absolute error has a narrower distribution than a normal distribution and the standard deviation highly depends on a small number of extreme outliers. The relative error is close to a normal distribution. Furthermore, the relative error decreases with increasing reflectivity and has a minimum around 33–35 dBZ (Figs. A3 and A4), suggesting that the relative error is sensitive to light and extreme precipitation. Since the maximum contribution to the total rainfall accumulation occurs around 35 dBZ (Fig. A4) and the smaller variability of DSDs is centered near that same reflectivity, there is less error in estimating rain accumulation than instantaneous rain intensity.

When using a climatological R - Z relationship, daily accumulations have a bias of 28% (Fig. 5). This bias could be reduced to 8% if the daily R - Z relationship were known. In addition to the bias, a random error of 32% in instantaneous rain estimation is expected due to the variability within a day. The improvement of the

random error from 41% (the climatological variability) to 32% (the variability within a day) is a consequence of taking into account the daily physical processes. Hence, it is concluded that significant variability comes from between storms within a day and classification of physical processes is necessary in order to reduce errors.

Based on the physical signature from the vertical structure of the precipitation, the classification of different physical processes is performed and then we have calculated the variability between the physical processes. As expected, the DSD variability between storms in which rain formation was produced by identifiably different physical processes is larger than the day-to-day variability. This variability yields a bias of 41% in rain accumulation for a process (Fig. 7). If the physical processes could be identified (either by synoptic information, multiparameter radar measurements, or from storm echo morphology), the error in rain estimation due to DSD variability could be reduced to a random error of 7%. Overall results suggest that the physically based classification of precipitation and the use of proper R - Z relationships are essential in order to reduce the effects of DSD variability in rain estimation.

An obvious question will be how we can possibly achieve this minimum random error of 7%. Recent research by the authors (Zawadzki and Lee 2004) reveals that there are reasonably well identifiable connections between R - Z relationships and the vertical structure of precipitation. For example, the riming process can be characterized by a weak bright band and relatively large vertical Doppler fall velocity ($>2\text{ m s}^{-1}$). It yields narrow DSDs due to the slow growth of bigger drops, resulting in a smaller exponent than that of the M-P R - Z relationship ($b = 1.5$). Contrary to riming, strong aggregation is characterized by a thick and intense BB due to fast growth of bigger drops, resulting in wide DSDs. Thus, the exponent of the R - Z relationship should be larger than 1.5. Scanning radars can provide a good vertical structure at near range. Therefore, investigating the use of the proper R - Z relationships in the various regions of the scanning radar coverage characterized by a predominant physical process are worthwhile. As reported by several researchers, polarimetric measurements can at least in theory be used to reduce the effect of the DSD variability in R estimation (Chandrasekar et al. 1990; Ryzhkov and Zrnić 1995; Brandes

TABLE 3. A summary of random and bias errors at various scales due to the variability of DSDs.

Scale of interest	R - Z relationship	Integration time	Bias (%)	Random (%)
Climatological scale	Climatological	Instantaneous R	28	41
		2-h accumulation		34
Day to day	Climatological	24-h accumulation	41	32
Within a day	Daily	Instantaneous R		
Between physical processes	Climatological	Accumulation for a process	7	
Within a physical process	Physical process	Instantaneous R		

et al. 2003). However, operationally, they suffer from large statistical measurement noise, making the use of pixel data impractical. A more realistic approach would be to average polarimetric data over larger areas in order to derive physical parameters. Since polarimetric parameters already contain information on drop shape, their availability should help in identifying the microphysical signature of precipitation regions.

Most of our DSD database comes from a single location: Montreal. Hence, the geographical variability is obviously not included in our analysis. Deep convective thunderstorms generate rain with DSDs that are very often closer to equilibrium between coalescence and breakup. Orographic and maritime rain may have different DSD characteristics. Certainly the drizzle observed here often follows the Marshall–Palmer behavior with large N_0 and narrow distribution but, equally as often, it is radically different from Hawaiian rain. However, as revealed by other investigators (Fujiwara 1965; Doelling et al. 1998), the difference in the R – Z relationship at different geographical locations is marginal except for thunderstorms (Florida versus Illinois; various locations in Europe). Thus, our conclusions might also broadly apply elsewhere, particularly in regions where stratiform precipitation dominates and the DSDs are shaped by processes occurring in snow above the melting layer.

Acknowledgments. This work was partially supported by a grant from the Canadian Foundation for Climate and Atmospheric Sciences. The authors are indebted to Dr. Aldo Bellon for numerous constructive comments and for editing this manuscript.

APPENDIX

Error Distributions in R Estimate Due to the DSD Variability at Climatological Scale

In this appendix, we now examine whether the errors in R estimation due to the DSD variability at the climatological scale are normally distributed. This investigation will provide a complete picture of the error structure in R estimation as well as a clue on whether the error distribution can be parameterized with a simple distribution model (i.e., normal distribution). Detailed error distributions are shown in Figs. A1 and A2 as a function of the absolute and relative error without and with SIFT, respectively. A comparison is done with the normal distribution (the dashed curve in the figures), which has the same mean and standard deviation as the actual distribution. Here, R_T indicates the estimated or transformed rainfall rate from reflectivity using our climatological relationship. The horizontal solid line shows the range of $\pm SD$ and the percentage of data within this range is provided above it.

Results show that the *absolute error* ($R - R_T$) (in Fig. A1a) is quite narrow and most of the errors are within one standard deviation (SD). The comparison with the normal distribution suggests the presence of extreme outliers. Since the absolute error has large weights on higher rain intensity points, this indicates that most outliers come from heavy rain. For the given dataset, the average rainfall intensity of outliers (only 3% of data), which are not within the range of $\pm SD$, is 23 mm h^{-1} . Hence, in terms of rainfall accumulation, the correct estimation of these extreme outliers is important in reducing the absolute error.

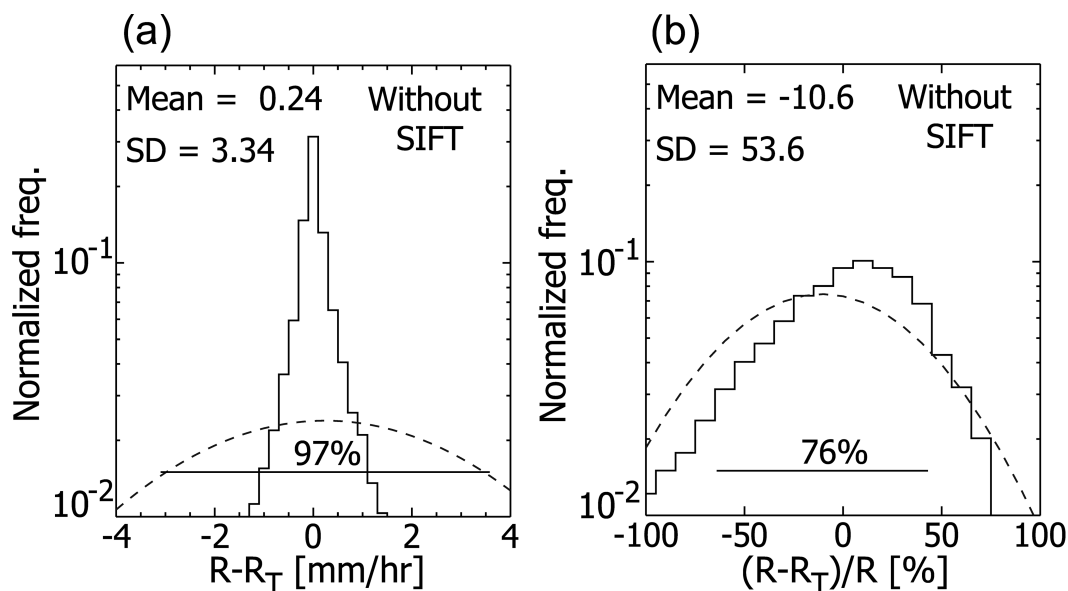


FIG. A1. Distribution of the absolute, $R - R_T$, and relative, $(R - R_T)/R$, errors in rain estimation with the climatological R – Z relationship. The dashed curve is the normal distribution that has the same std dev (SD) and mean as the actual histogram. The horizontal solid line indicates the range of $\pm SD$, and the percentage of cases within $\pm SD$ is given by the number above the line.

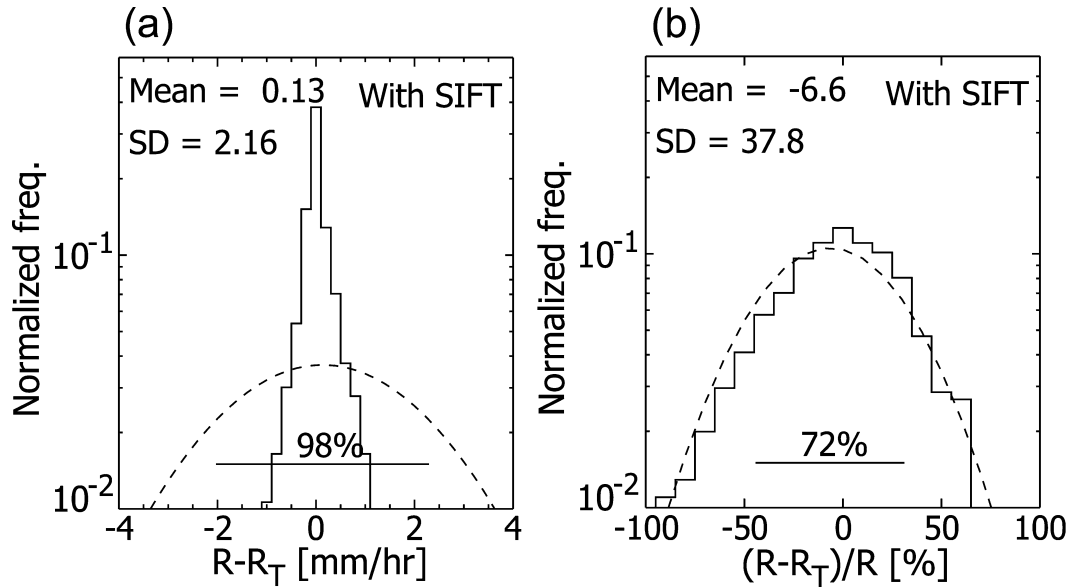
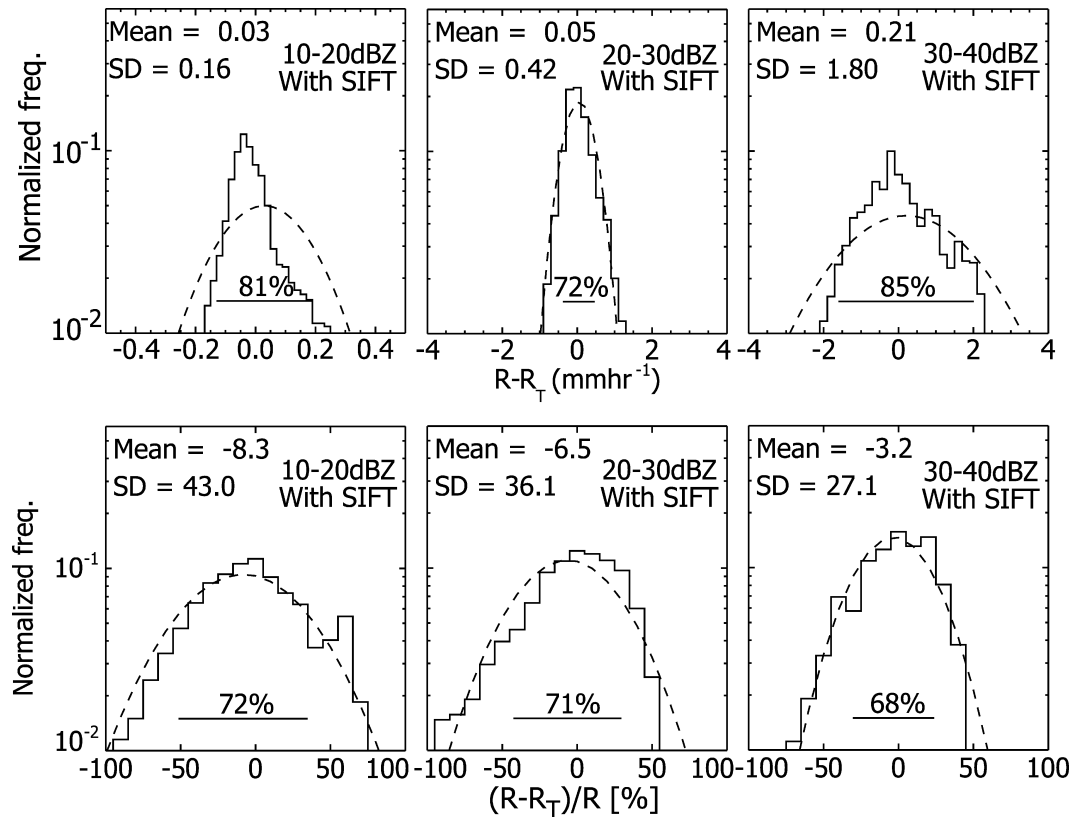


FIG. A2. Same as in Fig. A1 but for applying SIFT as in Fig. 2b.

FIG. A3. Distribution of the (top) absolute and (bottom) relative errors in rain estimation from Z for three different intervals of reflectivity. SIFT is applied as in Fig. 2b. The normal distribution and the range of $\pm\text{SD}$ are indicated as in Fig. A1.

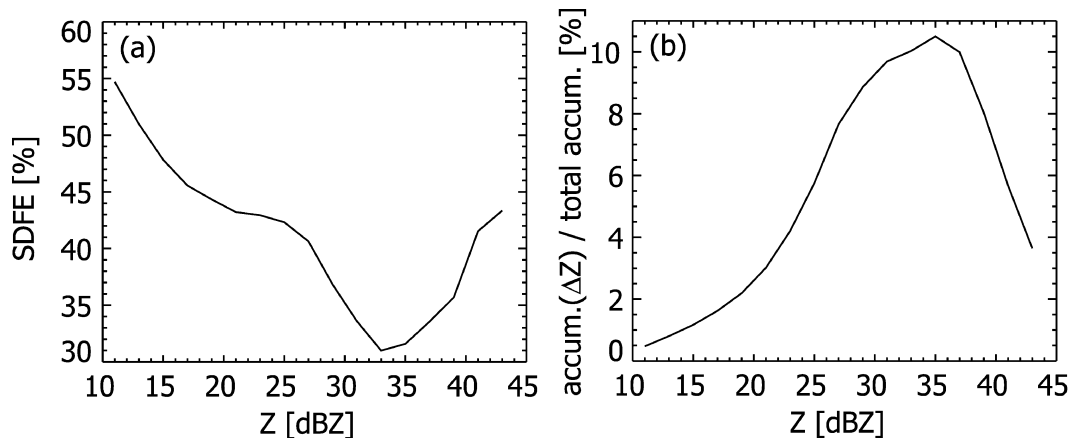


FIG. A4. (a) SDFE in the estimation of R using the climatological R - Z relationship ($Z = 210R^{1.47}$). The data in Fig. 2b are used. (b) Contribution to the total rain accumulation of different rain intensities within 2-dBZ intervals.

On the other hand, the distribution of the *relative errors* $(R - R_T)/R$, and of points found within a standard deviation, is closer to the normal distribution (in Fig. A1b). Moreover, the distribution of the absolute and relative errors is slightly skewed with positive and negative mean biases, respectively. The ratio of the mean bias of the absolute and relative errors with the SD at about 7% and 20%, respectively. After applying SIFT (Fig. A2), the error distribution becomes narrower, the outliers are less extreme, and the SD and mean bias are reduced. But the overall characteristics remain. As explained in section 2a, SIFT eliminates some of the instrumental noise and observational error. Therefore, when the R - Z climatological relationship is applied in radar rain estimations, the error distribution as shown in Fig. A2 will provide a base for the estimation error due to the variability of DSDs.

We now perform a similar analysis with the data stratified into 10-dBZ reflectivity intervals. The interval $Z > 40$ dBZ is not considered due to an insufficient sample size. Only the results with SIFT are shown in Fig. A3. The absolute error is far from a normal distribution except for the interval, $20 < Z < 30$ dBZ. The SD and mean bias increase with reflectivity. The distribution of the relative error is close to the normal distribution and broadens with decreasing reflectivity. When rain intensities are estimated from $30 < Z < 40$ dBZ, the SD of relative error is only 27% (much less than the overall random error of 41% in Fig. 2b). The mean bias is also small in the higher reflectivity range. To investigate the variation of the random error as a function of reflectivity factor, we calculate the standard deviation of fractional error (SDFE) within 5-dBZ intervals (centered at 2-dBZ steps from 11 to 43 dBZ) from the scatterplot in Fig. 2b using the climatological R - Z relationship ($Z = 210R^{1.47}$). The SDFE decreases with increasing reflectivity with a minimum around 33 dBZ to 35 dBZ (Fig. A4a). This illustrates that the relative error in instantaneous rain intensity is caused

by mostly light precipitation and to a lesser extent by extremely high precipitation. In addition, the contribution to the total rain accumulation of intensities within 2-dBZ intervals, shown in Fig. A4b, has a maximum around 35 dBZ ($\sim 7 \text{ mm h}^{-1}$). It is fortunate that the intensities that contribute most to the total have the smallest errors! Hence, for rain accumulation during a long period, the errors in the estimate of intensities in the 30–40-dBZ interval are the most relevant, in particular the errors due to DSD variability. For example, daily accumulations that are affected by the range of the maximum contribution should have less uncertainty than instantaneous rain intensity, since rain in the 30–40-dBZ interval contributes to daily accumulation significantly. This may be one reason that Joss and Waldvogel (1970) have an error of 13% in daily accumulation by using only three different R - Z relationships.

REFERENCES

- Amemiya, Y., 1997: Generalization of the TLS approach in the errors-in-variables problem. *Recent Advances in Total Least Squares Techniques and Errors-in-Variables Modeling*, S. Van Huffel, Ed., SIAM, 77–86.
- Balakrishnan, N., D. S. Zrnić, J. Goldhirsh, and J. Rowland, 1989: Comparison of simulated rain rates from disdrometer data employing polarimetric radar algorithms. *J. Atmos. Oceanic Technol.*, **6**, 476–486.
- Bell, C., 2000: Detection of the riming process with a vertically pointing radar. M.S. thesis, Dept. of Atmospheric and Oceanic Sciences, McGill University, 74 pp.
- Brandes, E. A., G. Zhang, and J. Vivekanand, 2003: An evaluation of a drop distribution-based polarimetric radar rainfall estimator. *J. Appl. Meteor.*, **42**, 652–660.
- Campos, E., and I. Zawadzki, 2000: Instrumental uncertainties in Z - R relations. *J. Appl. Meteor.*, **39**, 1088–1102.
- Chandrasekar, V., and V. N. Bringi, 1987: Simulation of radar reflectivity and surface measurements of rainfall. *J. Atmos. Oceanic Technol.*, **4**, 464–478.
- , —, N. Balakrishnan, and D. S. Zrnić, 1990: Error structure of multiparameter radar and surface measurements of rainfall. Part III: Specific differential phase. *J. Atmos. Oceanic Technol.*, **7**, 621–629.

- Direskeneli, H., K. Aydin, and T. A. Seliga, 1986: Radar estimation of rainfall rate using reflectivity and differential reflectivity measurements obtained during Maypole '84: Comparison with ground-based raingauges. Preprints, *23d Conf. on Radar Meteorology*, Snowmass, CO, Amer. Meteor. Soc., 116–120.
- Doelling, I. G., J. Joss, and J. Riedl, 1998: Systematic variation of Z-R relationships from drop size distributions measured in northern Germany during seven years. *Atmos. Res.*, **47–48**, 635–649.
- Fabry, F., and I. Zawadzki, 1995: Long-term radar observations of the melting layer of precipitation and their interpretation. *J. Atmos. Sci.*, **52**, 838–851.
- Fujiwara, M., 1965: Raindrop-size distribution from individual storms. *J. Atmos. Sci.*, **22**, 585–591.
- Goddard, J. W. F., and S. M. Cherry, 1984: The ability of dual polarization radar (co-polar linear) to predict rainfall rate and microwave attenuation. *Radio Sci.*, **19**, 201–208.
- Gunn, R., and G. D. Kinzer, 1949: The terminal velocity of fall for water droplets in stagnant air. *J. Meteor.*, **6**, 243–248.
- Hu, Z., and R. C. Srivastava, 1995: Evolution of raindrop size distribution by coalescence, breakup, and evaporation: Theory and observations. *J. Atmos. Sci.*, **52**, 1761–1783.
- Joss, J., and A. Waldvogel, 1970: A method to improve the accuracy of radar measured amounts of precipitation. *Proc. 14th Conf. on Radar Meteorology*, Tucson, AZ, Amer. Meteor. Soc., 237–238.
- , and I. Zawadzki, 1997: Raindrop size distribution again? Preprints, *28th Conf. on Radar Meteorology*, Austin, TX, Amer. Meteor. Soc., 326–327.
- Lee, G. W., 2003: Error in rain estimation by radar: Effect of the variability of drop size distributions. Ph.D. thesis, McGill University, 279 pp. [Available online at http://www.radar.mcgill.ca/~gwlee/PAPERS/THESIS_2003/thesis_gwlee_single.pdf]
- , and I. Zawadzki, 2003: Sequential intensity filtering technique (SIFT): Filtering out noise to highlight the physical variability of drop size distributions. Preprints, *31st Conf. on Radar Meteorology*, Seattle, WA, Amer. Meteor. Soc., 18–21.
- , and —, 2004: Radar calibration by gage, disdrometer, and polarimetry: Theoretical limit caused by the variability of drop size distribution and application to fast scanning operational radar data. *J. Hydrol.*, in press.
- , and —, 2005: Variability of drop size distributions: Noise and noise filtering in disdrometric data. *J. Appl. Meteor.*, in press.
- , W. Szymer, D. Sempere-Torres, and R. Uijlenhoet, 2004: A general approach to double-moment normalization of drop size distributions. *J. Appl. Meteor.*, **43**, 264–281.
- Marshall, J. S., and W. McK. Palmer, 1948: The distribution of raindrops with size. *J. Meteor.*, **5**, 165–166.
- Mitchell, D. L., 1988: Evolution of snow-size spectra in cyclonic storms. Part I: Snow growth by vapor deposition and aggregation. *J. Atmos. Sci.*, **45**, 3431–3451.
- Passarelli, R. E., Jr., 1978: Theoretical and observational study of snow-size spectra and snowflake aggregation efficiencies. *J. Atmos. Sci.*, **35**, 882–889.
- , and R. C. Srivastava, 1979: A new aspect of snowflake aggregation theory. *J. Atmos. Sci.*, **36**, 484–493.
- Richards, W. G., and C. L. Crozier, 1983: Precipitation measurement with a C-band weather radar in southern Ontario. *Atmos.–Ocean*, **21**, 125–137.
- Rogers, R. R., I. I. Zawadzki, and E. E. Gossard, 1991: Variation with altitude of the drop-size distribution in steady light rain. *Quart. J. Roy. Meteor. Soc.*, **117**, 1341–1369.
- Ryzhkov, A. V., and D. S. Zrnić, 1995: Comparison of dual-polarization radar estimators of rain. *J. Atmos. Oceanic Technol.*, **12**, 249–256.
- Sempere-Torres, D., R. Sánchez-Diezma, I. Zawadzki, and J.-D. Creutin, 1999: DSD identification following a pre-classification of rainfall type from radar analysis. Preprints, *29th Conf. on Radar Meteorology*, Montreal, QC, Canada, Amer. Meteor. Soc., 632–635.
- Sheppard, B. E., 1990: Measurement of raindrop size distribution using a small Doppler radar. *J. Atmos. Oceanic Technol.*, **7**, 255–268.
- , and P. I. Joe, 1994: Comparison of raindrop size distribution measurements by a Joss–Waldvogel disdrometer, a PMS 2DG spectrometer, and a POSS Doppler radar. *J. Atmos. Oceanic Technol.*, **11**, 874–887.
- Smith, P. L., and J. Joss, 1997: Use of a fixed exponent in “adjustable” Z–R relationships. Preprints, *28th Conf. on Radar Meteorology*, Austin, TX, Amer. Meteor. Soc., 254–255.
- , L. Zhong, and J. Joss, 1993: A study of sampling-variability effects in raindrop size observations. *J. Appl. Meteor.*, **32**, 1259–1269.
- Srivastava, R. C., 1967: On the role of coalescence between raindrops in shaping their size distribution. *J. Atmos. Sci.*, **24**, 287–292.
- Tokay, A., and D. A. Short, 1996: Evidence from tropical raindrop spectra of the origin of rain from stratiform versus convective clouds. *J. Appl. Meteor.*, **35**, 355–371.
- Uijlenhoet, R., J. A. Smith, and M. Steiner, 2003a: The microphysical structure of extreme precipitation as inferred from ground-based raindrop spectra. *J. Atmos. Sci.*, **60**, 1220–1238.
- , M. Steiner, and J. A. Smith, 2003b: Variability of raindrop size distributions in a squall line and implications for radar rainfall estimation. *J. Hydrometeorol.*, **4**, 43–61.
- Valdez, M. P., and K. C. Young, 1985: Number fluxes in equilibrium raindrop populations: A Markov chain analysis. *J. Atmos. Sci.*, **42**, 1024–1036.
- Willis, P. T., 1984: Functional fits to some observed drop size distributions and parameterization of rain. *J. Atmos. Sci.*, **41**, 1648–1661.
- Young, K. C., 1975: The evolution of drop spectra due to condensation, coalescence and breakup. *J. Atmos. Sci.*, **32**, 965–973.
- Zawadzki, I., and G. W. Lee, 2004: The physical causes of the variability of drop size distributions. Preprints, *14th Int. Conf. on Clouds and Precipitation*, Bologna, Italy, ICCP, 698–701.
- , E. Monteiro, and F. Fabry, 1994: The development of drop size distributions in light rain. *J. Atmos. Sci.*, **51**, 1100–1114.
- , F. Fabry, and W. Szymer, 2001: Observations of supercooled water and secondary ice generation by a vertically pointing X-band Doppler radar. *Atmos. Res.*, **59–60**, 343–359.